Academic Statement of Purpose

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In today's information age, the amount of long-form documents, articles, and other textual data offering high-quality information is increasing at an unprecedented rate. To grasp the overwhelming amount of content and extract the relevant content, an individual may need to expend significant time and cognitive effort. Furthermore, some of these resources include domain-specific knowledge semantics and technical concepts that may be incomprehensible to laypeople. My experience working on several NLP systems to improve document accessibility has shaped my research interests and instilled a desire in me to develop long document text processing systems that **eliminate knowledge barriers while enabling rapid document consumption**. To that end, I want to pursue a Ph.D. in Natural Language Processing (NLP) to deepen my understanding, solidify my foundation, and develop the skills required to lead focused research in this area.

My Excursion into the Legal Domain: Legal documents such as terms and conditions, agreements, etc., are ubiquitous. However, because of their notoriously complex linguistic structure and semantics, they are often incomprehensible to laypeople. Even if legal practitioners have the necessary knowledge to comprehend their intricacies, it takes a significant amount of time and effort to perform legal tasks such as risk analysis, contract review, legal question answering, summarization, etc. At Adobe Research, I had the unique opportunity to design an NLP system that processes a multi-page input contract to identify rights and responsibilities, link them with concerned parties, and flag potentially risky clauses. Not only would this reduce the burden of contract review, but it would also make a legal document more accessible to a layperson by presenting structured, multi-faceted information. This project introduced me to various information extraction techniques, assisted me in developing a critical eye to evaluate the usefulness of a solution, and was instrumental in shaping my research trajectory.

Working on this project made me realize the importance of legal language models and how they can be used to power the components of a legal system. Traditionally, these language models were pre-trained over legal corpora using MLM/Auto-regression without explicitly utilizing domain-specific characteristics. However, in my opinion, leveraging the domain knowledge and semantics during pre-training may furnish a superior language model with better downstream performance. To validate this hypothesis, I proposed a novel legal language model framework [4] that employed multi-task learning to understand different legal facets while pretraining. This approach yielded SoTA performance across several downstream tasks, which initiated plans to utilize it to power key features in **Adobe Sign**. While leading this research exploration, I understood the various stages of a project's life cycle, learned how to make a sound inductive hypothesis, and sharpened the necessary soft skills to articulate my research ideas and collaborate productively. My research endeavors have led to filing of **three patents** in this domain. In the future, I would like to extend this approach to other domains, such as health care, finance, etc., to expedite the development of resources that make specialty domain literature more accessible.

Towards Efficient Consumption for Long Documents: While my endeavors in legal AI focussed on overcoming knowledge barriers to accessing specialized resources, I was concurrently approaching the problem of improving the consumption experience of long documents - which are frequently time-consuming and tiresome to read regardless of how straightforward their content may be. A key observation highlights that they provide extensive information on various topics that may not be relevant to a specific user. Developing technologies that present pertinent content based on a user's topical interests can be a step toward achieving an efficient and personalized consumption experience. To accomplish this step, I worked on automatically **extracting the latent topical structure of a document text**. While most previous approaches model this as a linear segmentation task, I developed a supervised method for recovering its topical organization as a hierarchy by utilizing tree-structure conditional random field [3]. Being better aligned with the widely accepted theory of discourse segmentation [1], our approach performed significantly better against the prior techniques for several linear segmentation datasets. This was the first time I witnessed the success of combining linguistic theory with the solution method, which inspires me to look for intriguing possibilities where NLP can be connected with other disciplines to construct better computational models.

At Adobe, I leveraged this technology to develop the feature of topic-based summarization, personalizing

input documents based on a user's topical interests, and intelligent navigation experiences to enhance document consumption experiences. Currently, I am developing a text segmentation framework that involves an auxiliary objective to infer the aspect associated with the identified segments. This modeling technique would provide complementary information to improve segmentation and help unveil the topical themes in the document.

Prior Research Experiences: My initiative in making information accessible to all predated these experiences. At Adobe's research internship, I created a corpus-level system that established various semantic relations between documents in a user-uploaded directory to help users get a holistic picture of their collection. Here, we addressed the problem of detecting version relations between documents to minimize the cognitive effort required to locate the most recent document version and determine how a specific copy has evolved over time. As the naive pairwise computation of version relation would be computationally demanding, we used Minhash-LSH to approximately identify version sets and proposed a sophisticated post-processing algorithm to rectify any potentially noisy results. This work bagged the **best-paper runners-up award** at WISE 2021 [2]. To reduce the knowledge barriers to reading a constituent document, I also tackled the problem of determining the prerequisite documents from the corpus while developing this system. My proposed solution presented the prerequisite relations as an interactive dependency graph, which revealed insightful information like the best reading order for comprehension. These projects not only introduced me to NLP but also showed me how effective it can be in achieving accessibility and reducing cognitive load.

Why Ph.D.?: Through my research experiences, I encountered the problem of improving document accessibility under diverse scenarios and learned various solution modeling techniques while utilizing domain characteristics and cross-disciplinary concepts. However, I must undergo rigorous training to broaden my technological knowledge and deepen my understanding through focused research to envision a generalizable solution that applies to many use cases. A Ph.D. in NLP would help me acquire in-depth knowledge and also allow me to collaborate with experts from different disciplines, such as linguistics and cognitive science, which is critical for achieving my objective.

After completing my Ph.D., I aspire to become a professor and make a significant impact in my desired research area. My experiences as a TA during my undergraduate studies and as a research internship mentor at Adobe for **5** undergraduate teams have solidified my decision to pursue this career path.

Why UMich?: UMich's dynamic cohort of researchers with diverse research interests makes it an ideal place for me to pursue Ph.D. I am interested in working with **Prof. Lu Wang**, **Prof. Joyce Chai** and **Prof. Rada Mihalcea**. My background in discourse segmentation and interest in using it to improve the readability of long documents aligns with **Prof. Lu Wang**'s research explorations in incorporating hierarchical discourse structure in long document summarization and extracting functional discourse structure of news events. I would also like to collaborate with her on developing text summarization techniques that generate outputs aligned with user provided constraints such as topic, complexity, etc. I am interested to work under the guidance of **Prof. Joyce Chai** on developing conversation agents that aids the users in comprehending documents and in assessing how effective language models are in understanding specialty corpora. Inspired by the work on generating socially-aware questions, I would like to collaborate with **Prof. Rada Mihalcea** in designing user-driven systems that incorporates social information of the user to improve the readability and consumption experience of documents.

In summary, I bring diverse research experiences, industry-sharpened technical and soft skills, and a desire to create impactful solutions that enhance users' lives. I look forward to the next milestone in my life – a Ph.D. in Computer Science from UMich.

References

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